

Generalization of Text Line Segmentation for HTR in Historical Documents - Supplementary Material

We present supplementary material for our paper “Generalization of Text Line Segmentation for HTR in Historical Documents.”

1 Variants of Hi-SAM for Line Segmentation

We evaluated three variants of Hi-SAM: Base (B), Large (L), and Huge (H). The Base model is the smallest, whereas the Huge model is the largest variant. These models were used in our initial Hi-SAM based experiments on text line segmentation. We then evaluated their segmentation performance by computing the mAP at different confidence score thresholds, as shown in Figure S1. As shown in the figure, the Huge (H) variant consistently achieved the highest mAP50 : 95 across the evaluated thresholds, while the Large (L) variant produced the lowest performance.

2 Few-shot

2.1 Finetuning on Mix

In the main paper, we analyzed the fine-tuning of the models on subsets of target domains consisting of 10 or 50 pages. Since YOLO is significantly more cost-effective to train, we present additional results for finetuning with 5 and 25 pages in this supplementary material. These results provide further insight into how the models perform and allow for a more accurate assessment of how much annotated data is required for a new target dataset.

Both Tables S1 and S2 have been extracted from the main paper (main paper Tables 4 and 5), showing mAP and CER scores for models pretrained on the Mix dataset. The additional rows, especially for the CER in Table S2, show that even with 5 pages the segmentation performance of YOLO is already noticeably increased.

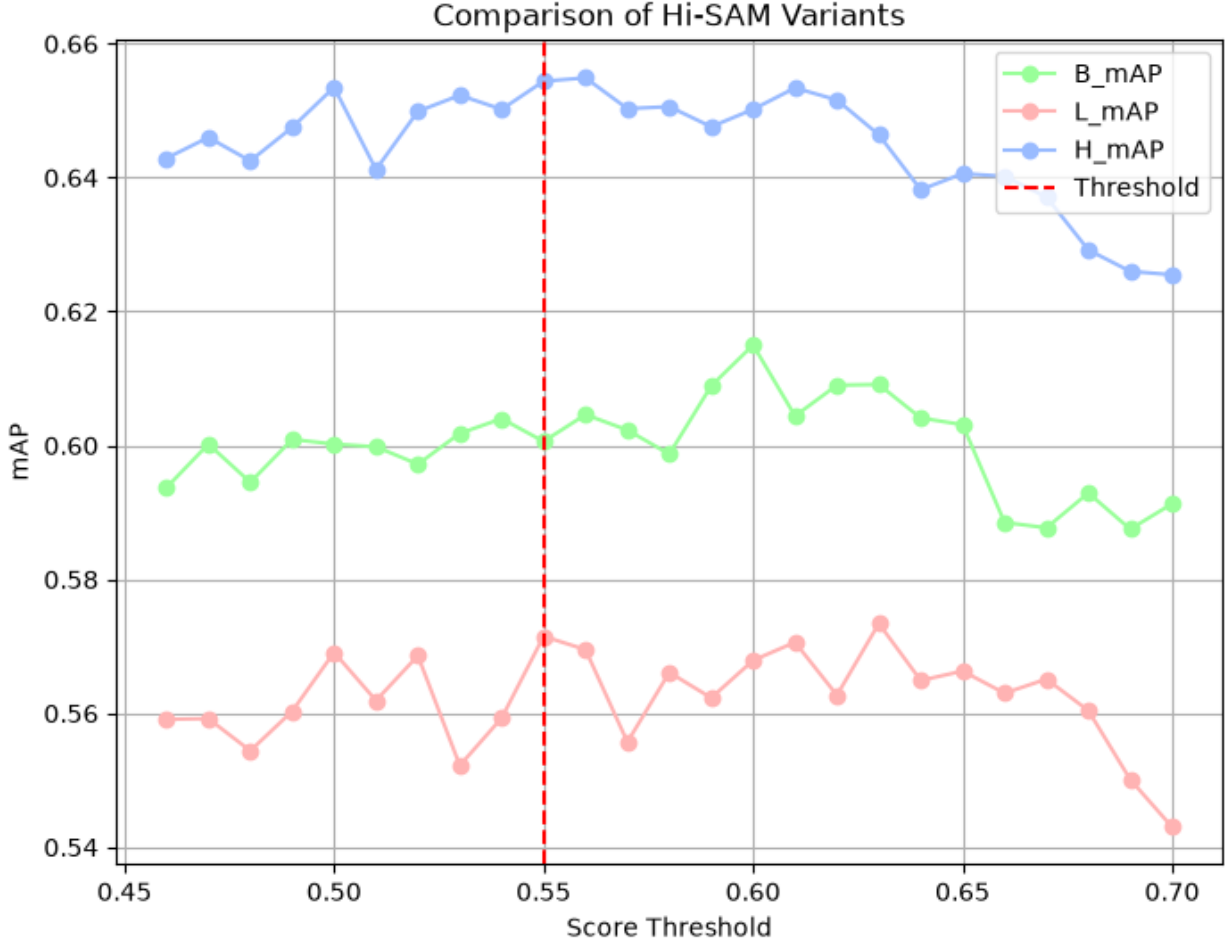


Figure S1: Segmentation performance of the Hi-SAM Base (B), Large (L), and Huge (H) variants at different confidence score thresholds, measured using $mAP_{50:95}$.

Table S1: Few-shot $mAP\%$ evaluation of Hi-SAM and YOLO line detectors pre-trained on the Mix dataset and fine-tuned on six target datasets. Higher values indicate better performance.

	Line Detector	Tested on					
		Bentham	NorHand 72702	Riksarkivet BK	Riksarkivet BK 2	Riksarkivet GBG 1	Riksarkivet GBG 2
Zero-shot	Hi-SAM	45.32	54.04	40.64	42.16	44.18	37.08
Few-shot 10	Hi-SAM	64.32	54.64	73.91	71.75	62.77	53.99
Few-shot 50	Hi-SAM	68.10	60.64	75.44	72.81	65.50	59.12
Zero-shot	YOLO	30.44	48.46	51.41	60.40	61.24	50.04
Few-shot 5	YOLO	61.21	57.35	80.57	79.13	63.67	58.05
Few-shot 10	YOLO	62.04	57.25	82.14	80.49	67.86	61.02
Few-shot 25	YOLO	68.48	62.03	82.09	82.05	68.95	65.09
Few-shot 50	YOLO	69.05	62.23	83.73	83.86	71.64	65.41

Table S2: Few-shot CER% evaluation of line detectors pre-trained on the Mix dataset and fine-tuned on six OOD datasets. Lower values indicate better performance. The GT row reports the performance of the HTR model when trained using ground-truth line bounding boxes.

	Line Detector	Tested on						OOD AVG gap with Baseline
		Bentham	NorHand 72702	Riksarkivet BK 1	Riksarkivet BK 2	Riksarkivet GBG 1	Riksarkivet GBG 2	
Baseline	GT-Boxes	2.22	1.61	3.40	2.48	4.60	2.96	
Zero-shot	Hi-SAM	4.44	3.36	4.20	3.14	7.77	11.76	2.90
Few-shot 10	Hi-SAM	3.31	2.72	3.89	2.64	6.59	8.25	1.69
Few-shot 50	Hi-SAM	3.22	2.56	3.57	2.69	6.50	5.89	1.19
Zero-shot	YOLO	5.48	10.81	4.09	3.67	7.53	11.87	4.36
Few-shot 5	YOLO	3.60	3.06	3.90	3.59	9.29	9.06	2.54
Few-shot 10	YOLO	3.79	3.08	3.80	2.93	7.12	8.80	2.04
Few-shot 25	YOLO	3.59	3.20	3.63	2.71	7.26	7.73	1.81
Few-shot 50	YOLO	3.64	2.79	3.49	2.48	7.07	6.32	1.42
COCO + Few-shot 5	YOLO	5.77	9.17	3.89	3.62	12.01	9.47	4.44
COCO + Few-shot 10	YOLO	6.12	94.56	3.75	16.03	8.27	9.93	20.23
COCO + Few-shot 25	YOLO	3.56	7.24	4.34	4.16	7.49	21.34	5.14
COCO + Few-shot 50	YOLO	4.31	2.65	3.55	2.55	7.13	5.60	1.42

2.2 Pre-Training on READ 2016

In this section, we first train the models with the READ 2016 dataset and fine-tune on the target out-of-domain (OOD) datasets. These experiments are similar to Tables 4 and 5 in the main paper, but the models are pre-trained with READ 2016 instead of the Mix dataset and the results are presented in Tables S3 and S4.

Table S3: Few-shot mAP% evaluation of three line detectors pre-trained on the READ 2016 dataset and fine-tuned on nine target datasets. Zero-shot rows report the corresponding results from Table 2 in the main paper; higher values indicate better performance.

	Line Detector	Tested on							
		Bentham	NorHand 72702	NorHand 331861	Riksarkivet BK 1	Riksarkivet BK 2	Riksarkivet GBG 1	Riksarkivet GBG 2	READ 2016 Nuremberg LB 3
Zero-shot	Hi-SAM	32.17	35.85	41.74	29.22	33.85	38.43	33.75	79.01 49.08
Few-shot 10	Hi-SAM	64.36	52.41	69.47	70.15	66.84	59.82	50.91	– 74.20
Few-shot 50	Hi-SAM	68.09	57.12	74.50	74.62	72.54	64.32	58.32	– 74.66
Zero-shot	YOLO	25.26	13.98	13.35	24.43	21.74	34.94	27.65	81.12 28.22
Few-shot 5	YOLO	58.95	52.53	74.26	80.59	76.43	61.44	53.30	– 71.01
Few-shot 10	YOLO	59.21	54.56	74.88	79.85	80.10	65.66	55.48	– 71.18
Few-shot 25	YOLO	65.59	58.37	76.13	81.34	81.06	67.80	62.08	– 72.09
Few-shot 50	YOLO	56.37	61.28	77.79	83.50	82.86	70.98	63.87	– 77.23

Table S4: Few-shot CER% evaluation of line detectors pre-trained on the READ 2016 dataset and fine-tuned on nine target datasets. Zero-shot rows report the corresponding results from Table 3 in the main paper; lower values indicate better performance. The GT row reports the performance of the HTR model when trained using ground-truth line bounding boxes.

	Line Detector	Tested on									OOD AVG gap with Baseline
		Bentham	NorHand 72702	NorHand 331861	Riksarkivet BK 1	Riksarkivet BK 2	Riksarkivet GBG 1	Riksarkivet GBG 2	READ 2016	Nuremberg LB 3	
Baseline	GT-Boxes	2.22	1.61	3.09	3.40	2.48	4.60	2.96	3.99	5.02	
Zero-shot	Hi-SAM	5.34	4.08	6.52	4.74	3.73	8.81	12.39	4.40	8.35	3.57
Few-shot 10	Hi-SAM	3.54	3.04	3.97	3.74	2.76	6.86	7.60	–	7.60	1.72
Few-shot 50	Hi-SAM	2.87	2.74	3.62	3.69	2.74	6.29	5.50	–	8.56	1.33
Zero-shot	YOLO	8.28	37.71	34.36	9.53	14.47	17.81	25.59	4.31	37.56	19.99
Few-shot 5	YOLO	6.16	4.01	4.09	3.77	3.87	10.16	13.12	–	8.34	2.56
Few-shot 10	YOLO	7.77	3.56	4.38	4.07	3.17	8.47	12.23	–	10.31	3.57
Few-shot 25	YOLO	5.46	3.17	4.66	3.73	2.77	8.41	9.10	–	10.10	2.00
Few-shot 50	YOLO	11.30	2.89	3.69	3.68	2.50	6.94	7.88	–	6.81	2.54

3 Impact of the Initialization Weights for Hi-SAM

In the main paper, the Hi-SAM experiments are presented using models initialized with SAM weights to ensure a fair comparison with YOLO and RT-DETR which were initialized on COCO, not document images. However, Hi-SAM can also be initialized using Hi-SAM pre-trained weights, which are document-aware and allow the model to converge faster.

Tables S5 and S6 present the results for models pre-trained on the Mix dataset. Using 10 few-shot samples improves the average out-of-domain CER by only 0.96 percentage points, indicating a marginal gain. Similar behavior is observed in Table 5 of the main paper, where the models were initialized with SAM weights. However, initializing the models with Hi-SAM weights significantly reduces the training time, as shown in Table 6 of the main paper.

Table S5: Few-shot mAP% evaluation of Hi-SAM **initialized with Hi-SAM weights**, pre-trained on the Mix dataset and fine-tuned on six target datasets. Zero-shot rows report the results without fine-tuning on the target datasets. Higher values indicate better performance.

	Line Detector	Tested on								
		Bentham	NorHand 72702	NorHand 331861	Riksarkivet BK 1	Riksarkivet BK 2	Riksarkivet GBG 1	Riksarkivet GBG 2	READ 2016	Nuremberg LB 3
Zero-shot	Hi-SAM	49.17	54.32	71.64	39.65	38.22	44.77	36.06	76.00	74.27
Few-shot 10	Hi-SAM	64.63	56.90	–	74.19	70.37	61.41	54.15	–	–
Few-shot 50	Hi-SAM	67.38	60.96	–	76.06	74.46	65.82	58.45	–	–

Table S6: Few-shot CER% evaluation of Hi-SAM **initialized with Hi-SAM weights**, pre-trained on the Mix dataset and fine-tuned on six target datasets. The GT row reports the performance of the HTR model when trained using ground-truth line bounding boxes. Zero-shot rows report the results without fine-tuning on the target datasets. Lower values indicate better performance.

	Line Detector	Tested on									AVG OOD CER gap with Baseline
		Bentham	NorHand 72702	NorHand 331861	Riksarkivet BK 1	Riksarkivet BK 2	Riksarkivet GBG 1	Riksarkivet GBG 2	READ 2016	Nuremberg LB 3	
Baseline	GT-Boxes	2.22	1.61	3.09	3.40	2.48	4.60	2.96	3.99	5.02	
Zero-shot	Hi-SAM	4.36	3.20	3.99	4.15	3.08	7.44	11.50	4.25	9.91	2.50
Few-shot 10	Hi-SAM	3.29	2.62	–	3.74	2.70	6.64	7.50	–	–	1.54
Few-shot 50	Hi-SAM	2.97	2.62	–	3.82	2.57	6.25	5.11	–	–	1.01

4 Qualitative Improvements of Few-shot 10

The qualitative samples in Figure 2 in the main paper show that Hi-SAM was the only model capable of accurately segmenting text lines in the zero-shot setting, while YOLO frequently produced incomplete line detections. However, after pre-training on the Mix dataset and fine-tuning with only 10 target-domain page samples, YOLO achieves a substantial improvement in segmentation quality, as shown in Figure S2. The predicted YOLO line boxes are now visually comparable to the ground-truth annotations.

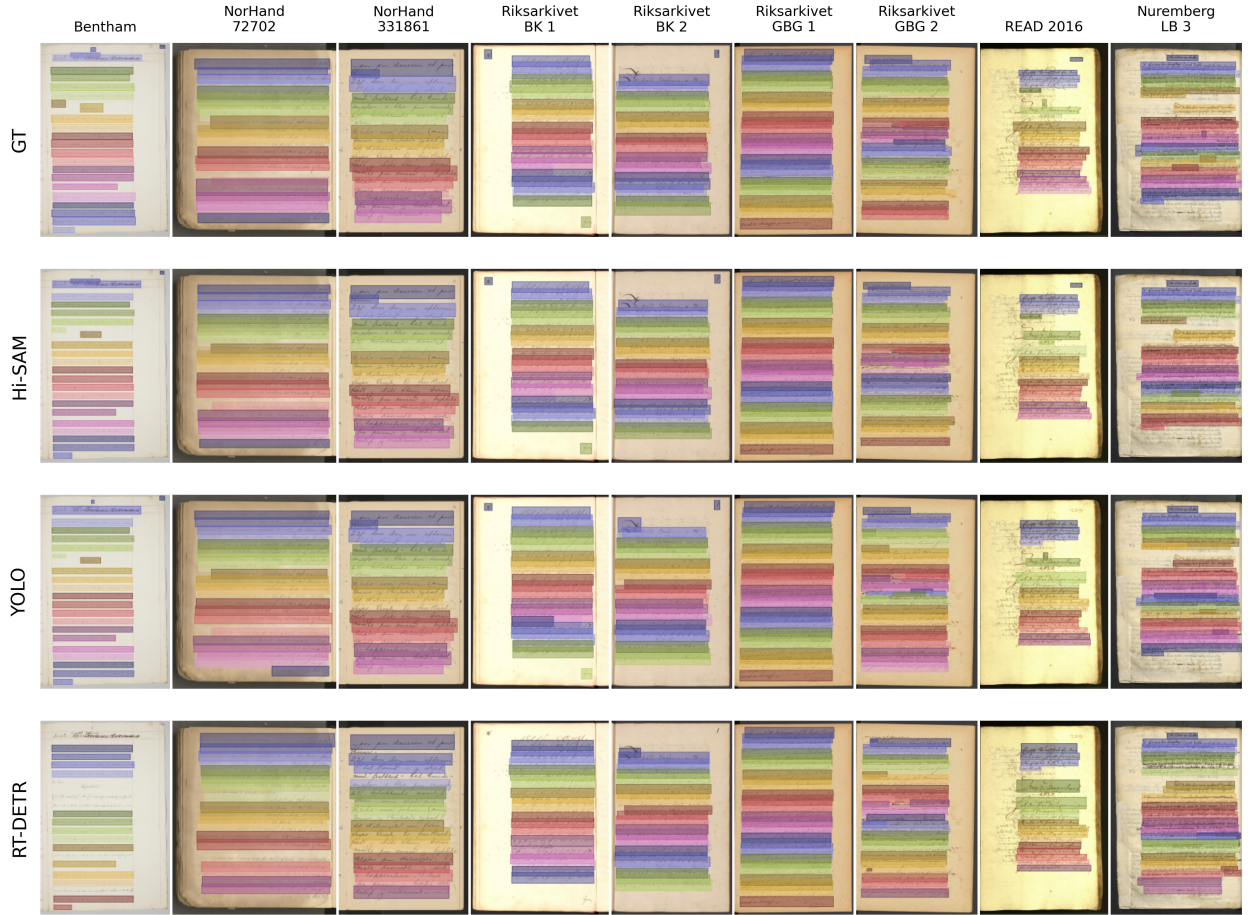


Figure S2: Qualitative comparison after pre-training on the Mix dataset and fine-tuning on 10 target dataset page images. The first row shows ground-truth boundary boxes; the remaining rows show the corresponding predictions from each line detection model on a samples from datasets we used in the study.